

Lokesh Reddy Sodanapalli

Embedded Systems Test & Validation Engineer

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Professional Experience

Test Automation Engineer, System Test – Working Student 01/2026 – Present

Joynext GmbH, Dresden, Germany

- Developed Python libraries and Robot Framework keywords for ECU testing using SSH and UART
- Automated RTCU lifecycle validation including power and low-power modes
- Performed debugging using DLT logs and Linux tools
- Worked with CAN, Automotive Ethernet, UDS, DoIP and TCP/IP

Software Test Engineer – Working Student

12/2024 – 07/2025

MBition GmbH, Berlin, Germany

- Executed system, integration and regression testing for infotainment domain controller
- Automated testing using Python and Robot Framework
- Performed network testing using Ethernet and TCP/IP
- Debugging and defect tracking using Jira

System Integration and Test Engineering Intern

05/2024 – 10/2024

Continental ADC GmbH, Ulm, Germany

- Built HIL-like test bench for radar sensors and ECUs
- Automated flashing and logging workflows using Python
- Reduced ECU CPU load from 90% to approx. 25%
- Conducted ADAS validation and calibration

Assistant System Engineer

06/2021 – 08/2022

Tata Consultancy Services, Bengaluru, India

- Infotainment validation for Jaguar Land Rover systems
- Functional and non-functional testing including Alexa features
- Test execution and defect tracking using Jira

Skills

Programming: Python, C, C++, Robot Framework, MATLAB/Simulink, Linux, Git

Tools: CANoe, CANdb++ , ODIS, Jira, Confluence, Jenkins, GitLab CI

Protocols: CAN, CAN-FD, Automotive Ethernet, TCP/IP, UDS, DoIP, UART

Domains: ECU Testing, ADAS Validation, Test Automation, Regression Testing, Debugging

Education

M.Sc. Embedded Systems

10/2022 – 05/2026

Technische Universität Chemnitz, Germany

Current Grade: 2.5

B.Tech Electrical & Electronics Engineering

07/2017 – 06/2021

Vellore Institute of Technology, India

Final Grade: 2.3

Projects

Validation of Ego-Motion Algorithm using Synthetic Radar Data – Master's Thesis

- Developed Python validation framework and evaluated velocity and yaw-rate using synthetic radar data

Home Energy Management System – Bachelor's Thesis

- Designed MATLAB/Simulink PV model and optimised battery utilisation

STM32 AC Signal Analysis – Master's Project

- Implemented DFT, DTFT and Goertzel algorithms for real-time DSP analysis

Intelligent Lighting System – Bachelor's Project

- Designed and implemented LDR-based lighting system with PCB development

Languages

English: C1

German: B1 (progressing to B2)